

Practical 9 CC Lab

```
%{
#include <stdio.h>
#include <stdlib.h>
int yylex();
int yyerror(char *s);
}%

%token NUMBER
%left '+' '-'
%left '*' '/'

%%

input : /* empty */
      | input line
      ;

line : expr '\n' { printf("Result = %d\n", $1); }
     | error '\n' { printf("Syntax Error! Please re-enter.\n"); yyerrorok; }
     ;

expr : expr '+' expr { $$ = $1 + $3; }
     | expr '-' expr { $$ = $1 - $3; }
     | expr '*' expr { $$ = $1 * $3; }
     | expr '/' expr { if ($3 == 0) { printf("Error: Division by zero!\n"); $$ = 0; } else $$ = $1 / $3; }
     | '(' expr ')' { $$ = $2; }
     | NUMBER
     ;

%%

int main() {
    printf("Desk Calculator: Enter expressions (Ctrl+C to exit)\n");
    yyparse();
    return 0;
}

int yyerror(char *s) {
    return 0; // Error message already printed in grammar
}

"practical9_1275.y" 37L, 810B
```

```
%{
#include "y.tab.h"
#include <stdlib.h>
}%

%%
[0-9]+      { yylval = atoi(yytext); return NUMBER; }
[+\\-*/\\n()] { return yytext[0]; }
.           { return yytext[0]; }

%%

int yywrap() {
    return 1;
}
```

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```
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim practical9_1275.l
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ vim practical9_1275.y
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ yacc -d practical9_1275.y
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ lex practical9_1275.l
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ gcc lex.yy.c y.tab.c -ll
lab-03-38@lab-03-38-OptiPlex-3280-AIO:~$ ./a.out
Desk Calculator: Enter expressions (Ctrl+C to exit)
1+2
Result = 3
6*5
Result = 30
4-2
Result = 2
8/2
Result = 4
□
```